IIoT and Cloud Computing (CS5005)

Lecture – 13

Network Layer

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, DESIGN AND MANUFACTURING, KANCHEEPURAM

Dr. Sanjeet Kumar Nayak

Assistant Professor

Department of Computer Sc. and Engg.

IP Adaptation for IoT Applications

- IP Layer
- IPv4
- IPv6

IP Adaptation for IoT Applications

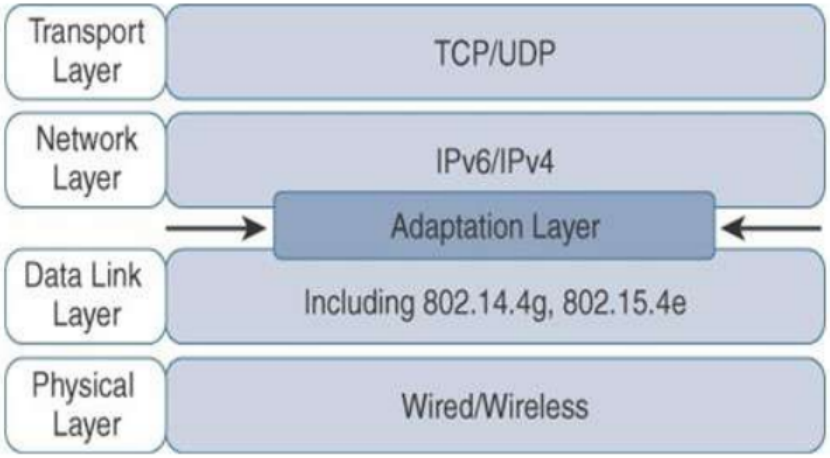
- Transport Layer
- UDP
- TCP

6LoWPAN

- 6LoWPAN: IPv**6** over **Lo**w-power **W**ireless **P**ersonal **A**rea **N**etworks

- IPv6 : header size (40 bytes) is too large for low-power networks
  - bandwidth and energy consumption are critical.

- IEEE 802.15.4 (Zigbee etc.)
  - Standard for low-power wireless communication
  - Provides small frame sizes (127 bytes maximum)
  - Low data rates (typically 250 kbps)
  - Direct IPv6 transmission inefficient.



6LoWPAN

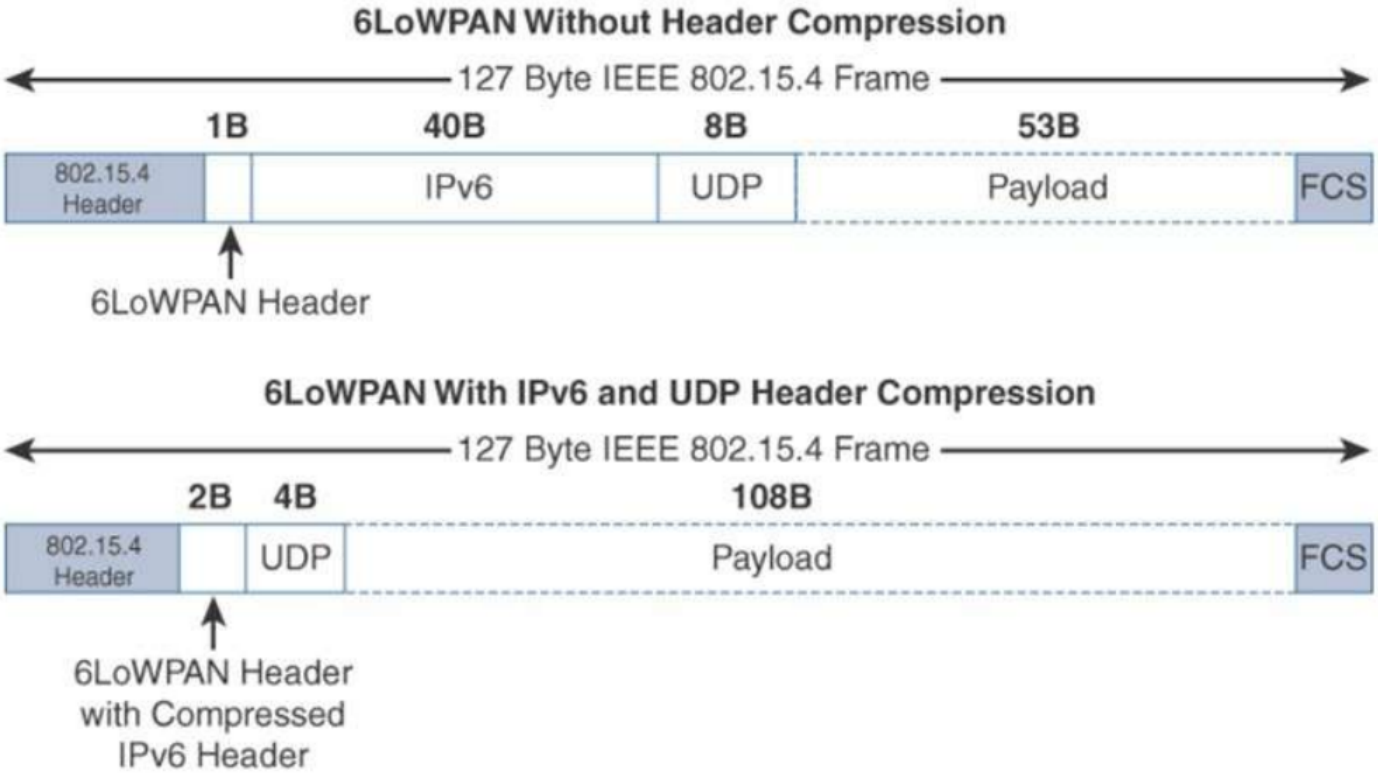
- 6LoWPAN header compression.

- Topologies:
  - Start
  - Mesh

- 6LoWPAN network architecture
  - Sensor Nodes
  - Routers
  - Gateway (links 6LoWPAN network and an IPv6, Convert compressed IPv6 packets back to full IPv6)
  - Cloud

6LoWPAN

- 6LoWPAN Header (Compressed) → 2 to 7 bytes (after compression)



Thank You

Any Questions?